

RUPAK PRADEEP POTDUKHE

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SUMMARY

Software Engineer with 2+ years of experience building scalable, consumer-facing platforms in **healthcare, hospitality, and AI**. Specialized in **JavaScript, React.js, Next.js, and TypeScript**, delivering high-performing, accessible, and responsive interfaces. Enhanced application speed and scalability for **50K+ daily users** while streamlining deployments with **Azure, Docker, and CI/CD** automation. Strong background in **API integration**, performance optimization, and test coverage with **Jest/React Testing Library** in **Agile** environments.

EDUCATION

Illinois Institute of Technology | USA

August 2023 – May 2025

Master of Science in Information Technology & Management

G H Raison College of Engineering | India

August 2019 – June 2023

Bachelor of Technology in Information Technology

PROFESSIONAL EXPERIENCE

Software Engineer | Virufy | USA

June 2024 – August 2024

- Engineered 12+ responsive UI components using **React (Hooks), Next.js, JavaScript, Tailwind CSS**, leveraging **AI-powered respiratory disease diagnostics** and supporting 2,000+ monthly user sessions.
- Owned the integration of **Python** backend services with MongoDB and **LLM APIs** into the React frontend, processing **JSON** payloads to deliver diagnostic updates in under 200 ms and powering a FAQ chatbot UI for patient queries across key workflows.
- Optimized application rendering performance using **Next.js** server-side rendering (**SSR**), image optimization, lazy loading, and memoization, cutting load time from 3.8s to 1.9s and improving **LCP, TTI**, and route transitions across 5 key user flows.
- Containerized frontend builds with **Docker**, and **Webpack 5**, deploying to Azure App Service via GitHub Actions CI/CD pipeline, automating weekly releases and eliminating setup errors across 3 sprints.
- Authored 30+ integration tests with **Jest** and **React Testing Library** for diagnostic workflows, validating **API contracts** through automated schema checks and cutting UI post-release defects from 14 to 2.
- Collaborated with stakeholders and designers via Figma and Jira, building 10+ WCAG-compliant UI components with semantic HTML, ARIA roles, and keyboard navigation, while instrumenting **Sentry** monitoring to track accessibility and runtime issues.

Software Engineer | Airbnb | India

August 2021 – July 2023

- Developed **14+ reusable UI modules** with **React.js, Redux, and TypeScript**, supporting 3,500+ weekly users and improving code maintainability across product teams.
- Integrated 20+ GraphQL and REST APIs backed by **SQL/NoSQL databases**, implementing **distributed** caching, concurrent request batching, and pagination strategies to reduce latency from 1.6s to 700ms while scaling to 45K+ requests/day.
- Spearheaded enhancements to Help Center search UI, architecting a **distributed** pipeline with Elasticsearch and Apache Kafka to process **concurrent** typeahead queries, scaling throughput to 45K+ requests/day and reducing search latency to 500ms.
- Resolved UI bottlenecks (redundant re-renders, hydration delays, excessive network fetches) and optimized performance with **Node.js SSR** and CDN strategies, reducing page load time from 4.2 s to 1.8 s across 50K+ daily sessions.
- Converted **10+ Figma prototypes** into production-ready, **WCAG 2.1-compliant UI** components, ensuring accessibility through axe audits and responsive testing while documenting patterns in Storybook for team-wide reuse.
- Strengthened frontend quality with **Jest** and **React Testing Library tests**, enforcing 85% coverage thresholds and gating CI builds tests/lint/types to reduce sprint bugs from 18 to 3.
- Collaborated in agile sprints with designers, PMs, and engineers; participated in code reviews and translated A/B test results and analytics (clicks, errors, conversions) into UI improvements aligned with stakeholder goals, reducing drop-off by 8%.

TECHNICAL SKILLS

Frontend: HTML5, CSS3, JavaScript (ES6+), TypeScript, React.js (Hooks, Router), Redux, Next.js, Tailwind CSS, SCSS.

Web & API Technologies: RESTful APIs, GraphQL, WebSockets, JSON, Swagger, Babel, Apache Kafka.

Backend & Databases: Node.js, Python, Express.js, MongoDB, PostgreSQL, MySQL.

Cloud & DevOps: AWS (EC2, S3, Lambda, API Gateway), Microsoft Azure (CI/CD), Docker, Jenkins.

Testing & QA: Pytest, Playwright, Jest, React Testing Library, Vitest, Postman, Lighthouse.

Monitoring & Security: Sentry, New Relic, OAuth 2.0, JWT, Passport.js.

Development Tools & Practices: Agile (Scrum), SDLC, Visual Studio Code, Git, GitHub, JIRA, Confluence, Figma (UI/UX).

PROJECTS

Jarvis AI – Smart Collaborative Workspace [link](#)

- Developed a real-time collaborative platform with **MERN stack** and **Google Gemini API**, allowing 10 users to chat, co-edit **AI-assisted code**, and execute it in-browser with synchronized updates, reducing setup time by 60%.
- Orchestrated a responsive **React (Hooks, JavaScript)** frontend to build UIs for integrated chat, collaborative code editing, and shared workspace, consuming real-time state via **WebSockets** and **AI-APIs**, achieving sub-50 ms latency while mitigating **XSS**.
- Architected scalable backend **APIs** with **Node.js** and **Express.js**, secured with **JWT authentication**, integrated **MongoDB** to store and retrieve AI-generated code, and validated **API-frontend integration** through **Jest testing** to ensure consistent functionality.

Movie Hub – Movie Streaming WebApp [link](#)

- Built a responsive front-end using **React.js (Hooks), Redux, Axios, and Tailwind CSS** to display 50+ movies and TV trailers, leveraging centralized state management to reduce re-renders by 60+ per session across desktop, mobile, and tablet.
- Configured a frontend **CI/CD** workflow with **Webpack, Babel, and Jest** on **Azure Cloud**, reducing deployment time by 3 hours per release while maintaining cross-browser compatibility and test coverage for core flows like search, playback, recommendations.